

Second Term Examination – 2082

Class : XI

Subject: C. English (0031)

Full Marks: 75

Time: 3 hrs.

SET B

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

1. Read the passage given below and answer the questions that follow.

Neanderthals and Modern Humans

A. The evolutionary processes that have made modern humans so different from other animals are hard to determine without the ability to examine human species that have not achieved similar things. However, in a scientific **masterpiece**, Svante Paabo and his colleagues from the Max Planck Institute for Evolutionary Anthropology, in Leipzig, have made such a comparison possible. In 2009, at a meeting of the American Association for the Advancement of Science, they made public an analysis of the genome of Neanderthal man.

B. Homo neanderthalensis, to give it its proper name, lived in Europe and parts of Asia from 400,000 years ago to 30,000 years ago. Towards the end of this period, it shared its range with interlopers in the form of Homo sapiens, who were spreading out from Africa. However, the two species did not settle down to a stable **cohabitation**. For reasons which are as yet unknown, the arrival of Homo sapiens in a region was always quickly followed by the disappearance of Neanderthals. Before 2009, Dr Paabo and his team had conducted only a superficial comparison between the DNA of Neanderthals and modern humans. Since then, they have performed a more thorough study and, in doing so, have shed a **fascinating** light on the **intertwined** history of the two species. That history turns out to be more intertwined than many had previously believed.

C. Dr. Paabo and his colleagues compared their Neanderthal genome (**painstakingly** reconstructed from three bone samples collected from a cave in Croatia) with that of five living humans from various parts of Africa and Eurasia. Previous genetic analysis, which had only examined DNA passed from mother to child in cellular structures called mitochondria, had suggested no interbreeding between Neanderthals and modern humans. The new, more extensive examination, which looks at DNA in the cell nucleus rather than in the mitochondria, shows this conclusion is wrong. By comparing the DNA in the cell nucleus of Africans (whose ancestors could not have crossbred with Neanderthals, since they did not overlap with them) and various Eurasians (whose ancestors could have crossbred

with Neanderthals), Dr Paabo has shown that Eurasians are between one percent and four percent Neanderthal.

D. That is intriguing. It shows that even after several hundred thousand years of separation, the two species were inter-fertile. It is strange, though, that no Neanderthal mitochondrial DNA has turned up in modern humans, since the usual pattern of invasion in historical times was for the invaders' males to mate with the invaded's females. One piece of self-knowledge, then - at least for non-Africans - is that they have a dash of Neanderthal in them. But Dr Paabo's work also **illuminates** the differences between the species. By comparing modern humans, Neanderthals, and chimpanzees, it is possible to distinguish genetic changes that are shared by several species of human in their evolution away from the great-ape lineage from those that are unique to Homo sapiens.

E. More than 90 percent of the 'human accelerated regions' that have been identified in modern people are found in Neanderthals too. However, the rest are not. Dr Paabo has identified 212 parts of the genome that seem to have undergone significant evolution since the species split. The state of genome science is still quite primitive, and it is often unclear what any given bit of DNA is actually doing. But an **examination** of the 20 largest regions of DNA that have evolved in this way shows that they include several genes that are associated with cognitive ability, and whose malfunction causes serious mental problems. These genes, therefore, look like good places to start the search for modern humanity's essence.

A. Mark the statements True, False, or Not Given. [5×1=5]

TRUE If the statement agrees with the information given in the text.

FALSE If the statement contradicts the information given in the text.

NOT GIVEN If there is no information in the text.

- a. Svante Paabo's team first published the full Neanderthal genome analysis in 2009.
- b. Neanderthals disappeared long before Homo sapiens entered Europe and Asia.
- c. Earlier mitochondrial DNA studies explained why Neanderthals became extinct after the arrival of modern humans.
- d. Modern Africans have no Neanderthal DNA in their genome.
- e. All human accelerated regions found in modern humans are also present in Neanderthals.

B. Choose among the highlighted words in the text that mean the same as the following definitions. [5×1=5]

- a. Existing together in the same place at the same time:
- b. Carefully and laboriously:
- c. Linked or closely connected:
- d. A work of outstanding artistry, skill, or workmanship:
- e. To clarify or explain something difficult:

C. Give short answers to the following questions in ONE SENTENCE each. [5×1=5]

- a. Why was Svante Paabo's research important for understanding human evolution?
- b. What happened to Neanderthals after Homo sapiens arrived in their regions?
- c. How did Dr Paabo's later DNA analysis differ from earlier studies?
- d. What percentage of Neanderthal DNA is found in modern Eurasians?
- e. What is the limitation of Genome science?

2. Write short answers to the following questions in about 75 words each. [5×2 = 10]

- a. What role does religion play in Aksionov's life? How does he undergo a spiritual transformation? (*God Sees the Truth but Waits*)
- b. Why do you think Mrs. Wrights asks Mrs. Peters for an apron in the prison? (*Trifles*)
- c. "Not everyone is capable of fulfilling the roles of the elders." Explain this statement with reference to the essay. (*Sharing Tradition*)
- d. Describe the oval portrait that the narrator saw in the chateau. (*The Oval Portrait*)
- e. Why is the last stage called second childhood? (*All the World's a Stage*)

3. Write long answers to the following questions in about 150 words each. [2×5 = 10]

- a. It is not easy to motivate others. How do you think Steve Jobs' speech is so inspiring? (*How to Live Before You Die*)
- b. How does the speaker express his love in 'A Red, Red Rose'? Do you think he is exaggerating? Why/not?

4. Write some paragraphs on 'The World 50 Years from Now' in about 150 words including title, topic sentences, and concluding sentences. [7]

5. Suppose your sibling is studying in the USA. Write a letter to him/her in around 180 words explaining the methods you have adopted to improve your grades. [8]

6. Draft a speech to be delivered on World Environment Day, highlighting the causes and effects of environmental pollution and suggesting ways to protect nature. (Word limit: 300 words) [10]

7. Do as directed and rewrite the sentences.

[10×1 = 10]

- a. I had an interesting dream yesterday. (*Identify the word class of the underlined word.*)
- b. My brother speaks fluently English. (*Place the underlined word in the right position.*)
- c. Ganesh almost fell.....the river. (*Complete the following sentence using 'to', 'toward', 'onto', or 'into'.*)
- d. Pritam(hide) behind the bush when he saw a teacher. (*Use the past form of the verb given in the brackets to complete the sentence.*)
- e. This exercise looks really hard. I..... help you. (*Complete the sentence using 'will' or 'be going to'.*)
- f. The jury..... (has/have) reached a unanimous decision. (*Complete the sentence using correct verb from the brackets.*)
- g. A number of students.....(is/are) playing outside. (*Complete the sentence using correct verb from the brackets.*)
- h. She dances very beautifully. (*Underline the verb in each sentence and write whether it is transitive, intransitive, or linking.*)
- i. They always.....(watch) horror films. (*Put the correct form of the word from the brackets to complete the sentence.*)
- j. You study now, or you will fail the exam. (*Put in 'had better' or 'should'.*)

8. Choose and copy the correct option.

[5×1 = 5]

- a. The word '**bickering**' means:
 - i. making a fool of someone
 - ii. arguing about insignificant things
 - iii. fighting over very important things
 - iv. shouting for help
- b. Which of the following words doesn't have any silent '**n**' letter?
 - i. solemn ii. autumn iii. damn iv. warns
- c. Find the odd one out:
 - i. inactive ii. input iii. interest iv. incomplete
- d. Choose the word with correct spelling.
 - i. apartheid ii. apartheid iii. aperthaid iv. aperteid
- e. Identify the word that ends with different sound.
 - i. looked ii. warned iii. washed iv. boost

☞ - The End - ☞